Extended Material: Simplifications are Absolutists: How Simplified Language Reduces Word Sense Awareness in LLM-Generated Definitions

Lukas Ellinger, Miriam Anschütz, and Georg Groh
School for Computation, Information and Technology
Technical University of Munich, Germany
{lukas.ellinger, miriam.anschuetz}@tum.de, grohg@cit.tum.de

A HoWN Construction

The Homonyms with WordNet (HoWN) dataset was constructed to evaluate LLM performance in generating definitions for homonyms with distinct senses. We filtered for nouns with at least two distinct definitions and more than two characters to exclude short, ambiguous terms. Nouns were chosen to avoid part-of-speech ambiguities; for example, words like *run* can function as both a noun (e.g., a jog) and a verb (e.g., to move quickly), with senses that are often closely related. Focusing on nouns ensured a clear separation of meanings, suitable for homonym definition tasks. This initial filtering yielded 1,780 candidate homonyms. To ensure these homonyms had multiple prevalent senses, rather than a dominant sense with rare secondary ones, we further filtered using the SemCor Corpus (Miller et al., 1993), the largest manually sense-annotated corpus based on WordNet, containing 226,040 sense annotations across 352 documents. We retained only words with at least two occurrences in SemCor, each annotated with a different sense, confirming their polysemy in real-world usage. This final step resulted in 164 target homonyms, each with verified, distinct senses suitable for evaluating LLM definition generation.

B Selected Models

We evaluated the following language models, selected for their diversity in size, architecture, and accessibility:

- **DeepSeek-V3** (DeepSeek-AI et al., 2025): An open-weight Mixture-of-Experts (MoE) model with 671 billion total parameters, activating 37 billion per token.
- **Llama 4 Maverick**[1]: A 400-billion-parameter MoE model with 128 experts, activating 17 billion parameters per token.
- **Qwen3-30B A3B** (Qwen Team, 2025): A 30-billion-parameter model from Alibaba's Qwen series, using an MoE architecture. We utilized its thinking mode for enhanced reasoning
- **GPT-4o mini**[2]: A lightweight variant of OpenAI's GPT-4o, optimized for faster response times and lower resource usage while maintaining competitive performance.
- **Llama 3.1 8B** (Grattafiori et al., 2024): An 8-billion-parameter model from Meta's Llama 3.1 series.

C Response Categorization

Figure 1 illustrates an example of a complete response, which includes all definitions (Full) as well as both key characteristics of Helpful Sense Awareness (HeSA): (i) a remark indicating that not all possible meanings may be covered, and (ii) a request for additional context.

D Automatic Evaluation

We found that splitting the categorization into two separate tasks, one for definition extraction and one for Helpful Sense Awareness (HeSA), improves evaluation accuracy. Additionally, we use a slightly adapted version of the definition extraction prompt for the ELI5 setting, as its stylistic characteristics differ from those of the Normal and Simple prompts. Figure 3 shows the system prompt for HeSA extraction. Figure 5 provides the system prompt for standard definition extraction. Figure 6 shows the adapted prompt for definition extraction

[1] https://huggingface.co/meta-llama/Llama-4-Maverick-17B-128E-Instruct

[2] gpt-4o-mini-2024-07-18, https://platform.openai.com/docs/models/gpt-4o-mini

The term "bat" can have multiple meanings depending on the context in which it is used. Here are the primary definitions:

- 1. Animal: A bat is a mammal belonging to the order Chiroptera. Bats are the only mammals capable of sustained flight. They are nocturnal and often associated with echolocation, which they use to navigate and hunt for insects in the dark.
- 2. Sports Equipment: In sports such as baseball or cricket, a bat is a specialized piece of equipment used to hit the ball. In baseball, it is typically a smooth, rounded wooden or metal club, while in cricket, it is a flat, paddle-shaped wooden implement.

If you have a specific context in mind, I'd be happy to help clarify!

Figure 1: Example response (adapted from DeepSeek v3). The green dashed line highlights the HeSA-style remark, while the blue underlined segment shows the context clarification request.

in the ELI5 setting. Finally, Figure 4 presents the user prompt layout used across all tasks.

E Direct Preference Optimization

We fine-tuned LLaMA 3.1 8B using Direct Preference Optimization (DPO) (Rafailov et al., 2023) to improve the completeness of homonym definitions. DPO trains a model on a dataset of prompt-response pairs, where each prompt is associated with a preferred (complete) and a non-preferred (incomplete) output. We constructed our training dataset using the HoWN dataset with the simple prompt type. For each response generated by LLaMA 3.1 8B, we evaluated its completeness. Incomplete responses were labeled as non-preferred, while complete responses from other models for the same prompt were labeled as preferred, yielding 116 preference pairs for 63 words. This process enhanced the model's ability to generate comprehensive definitions for homonyms. We provide fine-tuning details in Table 1.

Parameter	Value
LoRA Configuration	
r	64
LoRA Alpha	16
LoRA Dropout	0.05
Target Modules	[q_proj, v_proj, k_proj, o_proj]
Bias	none
DPO Training Configuration	
β	0.1
Learning Rate	5e-5
Batch Size (per device)	4
Epochs	10

Table 1: Combined configuration used for LoRA adaptation and Direct Preference Optimization (DPO) fine-tuning.

F Human Evaluation

To assess the reliability of our automated evaluation using an LLM-Judge, we conducted a human evaluation on the HoWN dataset. We randomly sampled 450 responses, with 150 responses per prompt type (ELI5, Simple, and Normal), annotated by one of the authors. We sampled responses from all models: Llama 4 Maverick (50 ELI5, 50 Normal, 50 Child), DeepSeek v3 (50 Simple), Qwen3-30B A3B (50 Normal), DPO Llama 3.1 (50 Normal, 50 Simple, 50 Child), Llama 3.1 (50 Simple), and GPT-4o mini (50 Child). We computed accuracy and Cohen's Kappa score to measure agreement between the human annotator and the framework for two tasks: Definition Count Classification (Category) and Helpful Sense Awareness (HeSA).

Table 2 reports the Cohen's Kappa scores and accuracy across prompt types. The combined Kappa scores are 0.86 for Category and 0.71 for HeSA, indicating almost perfect and substantial agreement, respectively (Landis and Koch, 1977). The average accuracy is 93.33% for Category and 86.44% for HeSA.

The variability in Kappa scores across individual prompt types stems from their distinct characteristics. For ELI5, most responses contain a single definition and lack HeSA. Conversely, Normal prompt responses exhibit a more balanced distribution of definitions and HeSA. This prompt-specific distribution affects the agreement scores, as Category and HeSA highly depend on the prompt type.

G Sense Coverage and Distribution Analysis

We analyzed which definitions most likely appear in model responses when not all coarse-grained synsets are covered. WordNet sense ranks are ordered by estimated frequency of use, with lower ranks corresponding to more commonly used

Prompt	Category	HeSA
ELI5	0.76 (93.33)	0.17 (94.67)
Simple	0.81 (92.00)	0.65 (84.67)
Normal	0.57 (94.67)	0.49 (80.00)
Combined	0.86 (93.33)	0.71 (86.44)

Table 2: Cohen's Kappa scores and accuracy percentages (in parentheses) for Definition Count Classification (Category) and Helpful Sense Awareness (HeSA) across prompt types on the HoWN dataset.

senses. We matched the extracted model definitions to their corresponding WordNet senses to enable this analysis. As shown in Figure 2, more prominent senses are included more frequently than rarer ones across all models and prompt types, which Based on our definition matching, we note that models occasionally describe the same WordNet sense more than once. As shown in Table 3, this occurs in an average of 56.51% of multi-definition responses for the Normal prompt, 31.29% for Simple, and 10.50% for ELI5. This difference can be attributed to fewer definitions being generated in simpler settings. The Normal prompt tends to be more fine-grained, often presenting the same sense in different contexts. As a result, the frequency of covered word senses remains relatively consistent across prompt types in Figure 2 since we count only whether a sense is covered, not how often.

Model	Normal	Simple	ELI5
Llama 3.1 8B	68.46	50.93	11.76
GPT-4o mini	52.74	21.28	0.00
Qwen3-30B A3B	62.32	25.40	7.84
Llama 4 Maverick	43.06	35.45	26.67
DeepSeek v3	56.03	23.40	6.25

Table 3: Duplicate WordNet Table.

H Multi-Sense-Aware Analysis

Table 4 extends the Table from the main section. Specifically, it includes additional metrics for *FKGL*, *Sense Awareness*, *Multiple Definitions*, *HeSA*, *Full*, and *Both*.

In the Normal setting, all models demonstrate perfect Sense Awareness by providing multiple definitions in every response, except for Qwen3-30B A3B, which falls just short with 99.32%. A similar pattern appears in the Simple setting, where Sense Awareness remains nearly perfect. GPT-4o mini achieves the lowest score at 95.27%, which is also the lowest among all models that provide multiple definitions. In the ELI5 setting, Sense Awareness also increases, with GPT-4o mini again at the lower end, reaching 85% Sense Awareness and 84.46% for multiple definitions. This aligns with expectations, as the system prompt explicitly encourages comprehensive explanations through the instruction:"Keep in mind that some words have more than one meaning." HeSA also improves significantly for the Simple and ELI5 prompts. For the Normal prompt, however, the impact varies by model. GPT-4o mini shows a substantial decline in HeSA, with a drop of 53.38 points, while Llama 4 Maverick demonstrates a significant improvement, gaining 22.30 points.

I ML-WIC Analysis

In Table 5, we present evaluation results on our ML-WIC dataset, including Sense Awareness, the proportion of responses with multiple definitions, and HeSA scores. Our DPO models consistently outperform all other models across every setting.

Table 6 presents the Flesch Reading Ease (FRE) score (Flesch, 1948), where higher values indicate simpler language. Scores are reported for English, French, and Russian, as FRE is unavailable for Arabic or Chinese. Across all models, the FRE score increases under simplification constraints, indicating that models adhere to the prompt instructions. We observe an exception with Llama 3.1 8B in the Normal setting for Russian, where the FRE score is negative. This is because the model sometimes produced flawed responses, often repeating words excessively. Additionally, we omit FRE scores for our DPO model, as it occasionally generated responses in English instead of the target language.

Table 7 reports the average number of definitions in responses that contain multiple definitions. A clear trend emerges: Normal yields more definitions on average than Simple, which in turn produces more than ELI5.

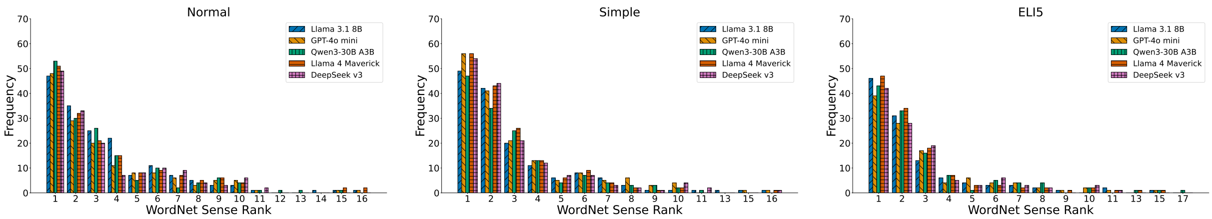

Figure 2: Frequency distribution of covered WordNet sense ranks across model-generated definitions. The x-axis represents the rank of WordNet senses, with lower ranks corresponding to more commonly used senses. The y-axis shows the frequency of occurrences for each rank, reflecting how often the model associates definitions with higher or lower-ranked senses. Only responses with partial coverage of coarse synsets were considered.

Model	FKGL	Sense Aware	Multi. Def	HeSA	Full	Both
Prompt: Normal						
Llama 3.1 8B	8.21 −0.65	**100.00** +2.70	**100.00** +2.70	60.81 −7.43	52.70 *+15.54*	29.73 +4.05
GPT-4o mini	8.62 −0.36	**100.00** +4.05	**100.00** +4.05	10.81 −53.38	56.08 *+6.08*	3.38 −30.41
Qwen3-30B A3B	6.51 −0.47	**100.00** +4.73	99.32 *+7.43*	**85.14** +4.73	**58.11** +12.84	**52.03** +12.16
Llama 4 Maverick	7.91 −1.00	**100.00** +4.05	**100.00** +4.05	82.43 *+22.30*	55.41 +8.78	41.89 *+15.54*
DeepSeek v3	7.52 −0.87	**100.00** *+6.08*	**100.00** +6.76	22.30 −1.35	66.22 +12.16	12.16 +4.05
Prompt: Simple						
Llama 3.1 8B	6.49 −0.34	97.30 +25.68	97.30 +26.35	15.54 −4.05	31.08 +13.51	4.73 −0.68
GPT-4o mini	6.31 +0.03	95.27 +32.43	95.27 +32.43	23.65 +16.22	38.51 +20.95	4.73 +2.03
Qwen3-30B A3B	5.18 −0.41	**100.00** +15.54	**100.00** +16.22	30.41 +8.11	**47.97** +16.89	12.16 +3.38
Llama 4 Maverick	6.13 −0.69	**100.00** +27.03	**100.00** +28.38	**60.14** *+33.11*	**47.97** *+27.70*	**30.41** *+21.62*
DeepSeek v3	5.89 −1.15	99.32 *+37.16*	99.32 *+37.16*	12.84 +7.43	47.30 +27.03	4.05 +4.05
Prompt: ELI5						
Llama 3.1 8B	2.77 −0.35	94.59 +80.41	93.92 +82.43	25.68 +21.62	22.30 +20.95	8.11 +7.43
GPT-4o mini	3.62 −0.33	85.14 +76.35	84.46 +75.68	7.43 +7.43	24.32 +22.97	1.35 +1.35
Qwen3-30B A3B	3.21 −0.58	**100.00** +62.84	**100.00** +65.54	**35.81** +25.68	34.46 +25.68	10.81 +8.11
Llama 4 Maverick	2.72 −0.26	96.62 *+86.49*	96.62 *+87.84*	33.11 *+30.41*	33.78 +33.11	**12.84** *+12.84*
DeepSeek v3	3.80 −0.32	97.30 +85.81	96.62 +85.81	19.59 +18.24	**35.81** *+35.14*	5.41 +5.41

Table 4: Performance metrics for models on the HoWN dataset under the Multi-Sense-Aware setting across Normal, Simple, and ELI5 prompt types. Metrics percentage of responses classified as Complete, Full (covering all meanings), HeSA (Helpful Sense Awareness), Both (Full and HeSA), and the average percentage of coarse-grained definitions covered. Deltas are reported to the non-Multi-Sense-Aware setting. coarse-grained definitions covered, with best scores for each prompt type in **bold**. The top value for each metric, within each prompt type and language, is highlighted in **bold**. The top delta is highlighted in *italic*.

Prompt / Model	Sense Aware					Multi. Def.					HeSA				
	En	Fr	Ar	Ru	Zh	En	Fr	Ar	Ru	Zh	En	Fr	Ar	Ru	Zh
Prompt: Normal															
Llama 3.1 8B	**96.97**	14.97	10.26	6.33	5.10	**95.96**	12.24	10.26	5.49	4.59	56.23	9.18	0.00	2.95	0.51
DPO Llama 3.1 8B	**99.66**	**99.32**	**99.15**	**99.16**	**98.47**	**99.66**	**98.30**	**98.29**	**97.89**	**96.94**	**80.13**	**95.24**	**86.32**	**94.51**	**53.06**
GPT-4o mini	93.94	78.91	92.31	90.30	84.69	93.94	77.89	**92.31**	89.45	83.67	28.62	17.35	17.09	27.85	16.33
Qwen3-30B A3B	94.61	86.39	**98.29**	89.87	**100.00**	93.60	84.01	91.45	86.50	**98.98**	**65.99**	**37.41**	**77.78**	**42.62**	**83.16**
Llama 4 Maverick	96.30	54.76	74.36	74.68	45.41	95.29	47.28	73.50	71.73	43.37	47.14	28.91	21.37	31.22	16.84
DeepSeek v3	94.28	**87.76**	91.45	**91.56**	87.76	93.60	**86.39**	91.45	**90.30**	86.22	20.88	23.13	30.77	18.57	46.94
Prompt: Simple															
Llama 3.1 8B	64.31	7.48	5.98	2.11	8.67	62.96	5.10	5.98	2.11	8.67	6.06	2.38	0.00	0.00	0.51
DPO Llama 3.1 8B	**92.93**	**93.54**	**96.58**	**99.58**	**93.88**	**92.93**	**89.12**	**96.58**	**98.31**	**93.37**	**54.88**	**69.39**	**61.54**	**73.84**	**31.63**
GPT-4o mini	63.30	53.06	76.07	43.88	75.51	63.30	53.06	76.07	43.46	75.51	2.69	0.00	0.00	3.38	1.02
Qwen3-30B A3B	**76.77**	**59.52**	69.23	**67.51**	**82.65**	**75.76**	**57.82**	65.81	**66.24**	**81.63**	11.45	**7.48**	**12.82**	**11.39**	24.49
Llama 4 Maverick	70.03	28.57	44.44	49.37	68.88	69.36	26.19	44.44	46.84	60.71	**15.15**	5.78	1.71	8.86	**26.53**
DeepSeek v3	63.97	47.96	**80.34**	65.40	73.98	63.97	47.28	**80.34**	64.14	73.47	2.02	1.70	4.27	8.86	13.27
Prompt: ELI5															
Llama 3.1 8B	7.07	7.48	0.85	1.27	0.51	6.73	3.40	0.00	0.42	0.51	0.67	4.42	0.85	0.84	0.00
DPO Llama 3.1 8B	**35.35**	**35.03**	**57.26**	**63.29**	**34.69**	**32.66**	**29.25**	**51.28**	**54.43**	**32.14**	**17.85**	**18.71**	**30.77**	**42.19**	**14.80**
GPT-4o mini	5.72	6.80	11.11	2.53	6.63	5.72	6.80	10.26	2.11	6.12	0.00	0.34	0.85	0.84	0.51
Qwen3-30B A3B	**22.22**	**17.35**	11.11	**14.35**	**14.29**	**19.87**	**11.56**	7.69	**13.08**	8.16	**4.71**	**8.16**	**4.27**	1.69	**6.63**
Llama 4 Maverick	10.77	12.93	11.97	9.70	9.69	8.42	10.20	11.97	7.59	6.63	3.70	5.10	0.85	**2.53**	3.06
DeepSeek v3	8.08	8.16	**12.82**	8.86	11.73	8.08	7.82	**12.82**	8.02	**11.73**	0.00	2.04	0.00	0.84	0.00

Table 5: Evaluation scores for Sense Awareness, Multiple Definitions, and HeSA across English (En), French (Fr), Arabic (Ar), Russian (Ru), and Chinese (Zh) for Normal, Simple, and ELI5 prompt types. The top two values for each metric, within each prompt type and language, are highlighted in **bold**.

Model	Normal			Simple			ELI5		
	En	Fr	Ru	En	Fr	Ru	En	Fr	Ru
Llama 3.1 8B	60.23	41.55	-15.62	75.33	75.59	53.07	95.12	92.75	60.16
GPT-4o mini	57.72	65.57	37.52	77.20	82.32	58.17	92.91	99.12	79.36
Qwen3-30B A3B	60.70	69.99	35.20	78.24	81.34	62.53	92.35	97.60	81.63
Llama 4 Maverick	58.27	61.06	31.13	71.41	65.70	49.44	95.05	94.28	76.11
DeepSeek v3	60.11	68.11	38.64	73.54	77.48	54.26	91.32	97.09	73.84

Table 6: Flesch Reading Ease (FRE) scores per prompt type (Normal, Simple, ELI5) across models and languages (English, French, Russian). Higher scores indicate easier-to-read text.

Model	Normal					Simple					ELI5				
	En	Fr	Ar	Ru	Zh	En	Fr	Ar	Ru	Zh	En	Fr	Ar	Ru	Zh
Llama 3.1 8B	4.80	4.28	2.58	3.85	2.33	2.86	2.13	2.00	2.20	3.35	2.25	2.20	0.00	2.00	3.00
DPO Llama 3.1 8B	5.06	4.95	3.73	5.26	3.67	3.82	3.50	3.19	3.97	3.60	2.81	2.36	2.38	2.97	2.59
GPT-4o mini	3.93	3.35	3.11	3.36	3.49	2.32	2.23	2.42	2.23	2.45	2.00	2.10	2.00	2.00	2.08
Qwen3-30B A3B	5.60	4.39	3.74	4.66	5.80	2.79	2.65	2.27	3.03	3.77	2.14	2.21	2.11	2.35	2.31
Llama 4 Maverick	4.46	3.63	6.84	3.86	3.40	2.84	2.49	2.81	2.68	3.13	2.12	2.03	2.07	2.22	2.23
DeepSeek v3	4.93	4.23	4.31	4.30	4.56	2.72	2.60	2.87	2.84	3.32	2.00	2.09	2.00	2.05	2.17

Table 7: Average number of definitions in responses that contain multiple definitions, reported per prompt type (Normal, Simple, ELI5) across models and languages.

```
You are a precise evaluator that determines two flags based on a response explaining a word definition.

Your task:

1. **remark_not_all_listed** -> Set this to True **only if** the response explicitly signals that the list of definitions is
     incomplete.
   - Set to True when the response includes phrases like:
     - "there are other meanings"
     - "this is just a selection"
     - "some examples include"
     - "among others"
     - "etc."
     - "this is not an exhaustive list"
   - Set to False when:
     - The response lists multiple meanings but gives **no signal** of incompleteness.
     - It introduces uncommon or surprising meanings **but does not suggest there are more**.
     - It uses grouping phrases like "has multiple meanings" or "depends on context" **to summarize the meanings given**,
          without implying that additional, unlisted meanings exist.
     - It uses playful or contextual comments like "that’s a grown-up thing" without signaling incompleteness.

2. **context_clarification_request** -> Set this to True **only if** the response explicitly asks the user to provide more
     context, clarification, or disambiguation.
   - Set to True when the response includes phrases like:
     - "if you have a specific context in mind, please tell me"
     - "can you specify the context the word is used in?"
   - Set to False when:
     - The response explains multiple meanings or contexts but **does not** request further clarification from the user.
     - It simply states that the word "depends on context" or "has multiple meanings" without asking for more input from the
          user.
     - It invites the user to interact or reflect (e.g., "Do you have any clips at home?"), but this interaction **is
          unrelated to clarifying the word’s meaning**.

**Example 1**:
Query: What is the definition of "bluff"?
Response: The term "bluff" has multiple definitions depending on the context in which it is used.
1. In geography or geology: ...
2. In poker: ...
3. In general usage: ...
The core idea involves deception or directness, depending on context. Does this make sense?

**Expected Output**:
{’remark_not_all_listed’: False, ’context_clarification_request’: False, ’explanation’: ’remark_not_all_listed is False
     because the response lists definitions without signaling incompleteness. context_clarification_request is False because
     no context is requested from the user.’}

**Example 2**:
Query: Explain me "clip" like I am 5 years old.
Response: Ooh, let me explain "clip" to you! ... There are different kinds of clips, like paper clips, hair clips, and even
     clips for bags or jackets. Isn’t that cool? Do you have any clips at home?

**Expected Output**:
{’remark_not_all_listed’: False, ’context_clarification_request’: False, ’explanation’: ’remark_not_all_listed is False
     because no signal of incompleteness is given. context_clarification_request is False because the interaction is playful,
      not a request for clarification.’}
```

Figure 3: System Prompt used to evaluate Helpful Sense Awareness

```
You are a helpful assistant. Answer the question as accurately as possible.

**Prompt**: "{query}"

**Model Response**: "{model_response}"
```

Figure 4: User prompt layout used in all requests.

```
You will extract definitions from the following response and determine whether it expresses one or multiple distinct meanings.

Instructions:

1. Read the response carefully.
2. Identify the distinct meanings or senses of the word being defined.

   - Treat it as **one definition** if the response elaborates, gives examples, describes subtypes, or mentions alternative
        forms **within the same core meaning**.

   - Split it into **multiple definitions** only if the response clearly and explicitly explains **separate meanings or senses
        ** that differ in their fundamental usage or context. This includes:
     - Different referents (e.g., object vs. concept)
     - Informal, colloquial, or figurative uses that shift the meaning
     - Literal vs. metaphorical senses
     - Different grammatical or functional roles (e.g., noun vs. verb)

   - Do **not** split definitions if:
     - There are multiple examples or elaborations
     - The response describes consequences, comparisons, or related features of the same concept
     - A second meaning is only hinted at or vaguely mentioned without being clearly defined

3. Provide three outputs:

   - 'definitions': a list of distinct definitions (if any).
   - 'category': either 'One' or 'Multiple'.
   - 'explanation': a brief reason for your categorization.

Example 1 (single meaning with details):

Input:
"March means to walk in a formal, organized way as part of a group, often with music and banners."

Output:
definitions:
- To walk in a formal, organized way as part of a group, often with music and banners.
category: One
explanation: This is a single meaning with elaborative details; no distinct senses are presented.

Example 2 (clear multiple meanings):

Input:
"Break can mean to separate into pieces by force, or to pause from an activity."

Output:
definitions:
- To separate into pieces by force.
- To pause from an activity.
category: Multiple
explanation: These are two clearly distinct senses of the word.

Example 3 (no split for subtype and mention):

Input:
"An ass is another kind of animal, also called a donkey. Sometimes people use the word in not-so-nice ways."

Output:
definitions:
- An animal, also called a donkey, similar to a horse but smaller with longer ears.
category: One
explanation: Only one meaning is fully defined; the other usage is mentioned but not explained as a separate sense.

Example 4 (no split for examples and types):

Input:
"A clip is a special tool that helps keep things together, like papers, hair, or bags."

Output:
definitions:
- A special tool that helps keep things together, like papers, hair, or bags.
category: One
explanation: These are examples and subtypes of the same core meaning.
```

Figure 5: System prompt used to extract definitions in the Simple and Normal prompting settings

```
You are a precise evaluator tasked with analyzing a model response defining a word in a child-friendly manner (e.g., prompted
      with 'Explain me [word] like I am 5 years old'). Your goal is to extract distinct definitions and categorize the
      response as having one or multiple meanings, recognizing that child-friendly responses often use imaginative or example-
      heavy language to describe a single core meaning.\n\n
Instructions:\n\n
1. **Read the Response Carefully**:
   - Identify the word being defined and all meanings or senses provided. Note that child-friendly responses often include
         metaphors, playful language, or multiple examples to engage young audiences.\n\n
2. **Extract Definitions**:
   - **One Definition**: Count as one if the response describes a single core meaning, even if it includes:
     - Imaginative metaphors (e.g., 'a big hug from the water' for a bay).
     - Multiple examples or subtypes (e.g., 'paper clips, hair clips' for a clip).
     - Playful or emotional elaborations (e.g., 'makes you feel like you can fly' for a cape).
     - Descriptions of uses, contexts, or feelings tied to the same concept (e.g., 'worn by superheroes or at Halloween' for a
            cape).
   - **Multiple Definitions**: Split into separate definitions only if the response **explicitly introduces distinct meanings
         or senses**, such as:
     - Different referents (e.g., a physical object vs. an abstract concept).
     - Literal vs. metaphorical senses with clear shifts in meaning.
     - Different grammatical roles (e.g., noun vs. verb) with distinct implications.
   - **Do Not Split**:
     - Examples, subtypes, or contexts that support the same core concept (e.g., 'playing soccer or with toys' for play).
     - Stylistic, emotional, or imaginative variations that do not change the core sense (e.g., 'a magic cloak' for a cape).
   - **Compound Sentences or 'or'-Phrases**: Evaluate carefully. Only split if they describe distinct senses with different
         implications (e.g., 'to put or adjust' may be two senses if contexts differ significantly).
   - **Kid-Friendly Language**: Treat metaphors, analogies, or engaging descriptions as elaborations of a single meaning
         unless they clearly introduce a new sense. For example, 'a big hug from the water' and 'calm place for boats'
         describe the same geographic feature (bay).
   - Do not verify factual correctness-extract only what is present.\n\n
3. **Assign Category**:
   - **None**: No definitions provided.
   - **One**: One distinct definition extracted.
   - **Multiple**: Two or more distinct definitions extracted.\n\n
4. **Provide Outputs**:
   - `definitions`: A list of distinct definitions, summarizing each sense concisely.
   - `category`: 'None', 'One', or 'Multiple'.
   - `explanation`: A brief reason for the categorization, addressing why details are treated as elaborations or distinct
         senses in the child-friendly context.\n\n
Example 1 (Single Meaning with Kid-Friendly Elaborations):
Input: 'A bay is a special area where the ocean curves into the land. It's like a big hug from the water! Bays are good places
       for boats because the water is calmer and protected from big ocean waves.'
Output:
{
  'definitions': ['A special area where the ocean curves into the land, providing a calm place protected from big waves'],
  'category': 'One',
  'explanation': 'The response describes one meaning of bay (a geographic feature) with kid-friendly metaphors ('big hug') and
         elaborations (calm water, boats). No distinct senses are introduced.'
}\n\n
Example 2 (Single Meaning with Playful Language):
Input: 'A cape is a special kind of clothing that you wear on your back! It's like a big piece of fabric that goes around your
       neck and hangs down behind you. Imagine you're a superhero, like Superman or Batman! You wear a cape to make you feel
      strong and brave. It's like a magic cloak that helps you fly (well, not really, but it makes you feel like you can!).
      Some people wear capes on Halloween or at costume parties to dress up like their favorite superheroes.'
Output:
{
  'definitions': ['A special kind of clothing that you wear on your back, like a big piece of fabric that goes around your
        neck and hangs down behind you'],
  'category': 'One',
  'explanation': 'The response defines cape as clothing with kid-friendly elaborations (superhero imagery, Halloween use,
        emotional effects). These are not distinct senses but support the same concept.'
}\n\n
Example 3 (Single Meaning with Examples):
Input: 'Ooh, let me explain "clip" to you! A clip is a special tool that helps us keep things together. It's like a little
      helper! You can use a clip to hold papers or keep hair in place. There are different kinds of clips, like paper clips or
       hair clips.'
Output:
{
  'definitions': ['A special tool that helps keep things together, like for papers or hair'],
  'category': 'One',
  'explanation': 'The response defines clip as a fastening tool with kid-friendly examples (papers, hair). These are
        elaborations of one meaning, not distinct senses.'
}\n\n
Example 4 (Multiple Meanings, Hypothetical):
Input: 'A bank is a place where you keep your money, like where your piggy bank savings go! It's also a piece of land next to
      a river, like where you might have a picnic by the water.'
Output:
{
  'definitions': ['A place where money is kept, like a financial institution', 'A piece of land next to a river'],
  'category': 'Multiple',
  'explanation': 'The response provides two distinct meanings of bank (financial institution and river land), each with kid-
        friendly descriptions. These are separate senses with different referents.'
}\n\n
Example 5 (Single Meaning with Cultural Context):
Input: 'Oh boy, are you going to love learning about the Cheyenne! So, a long, long time ago, there was a group of Native
      American people called the Cheyenne. They lived in a place called the Great Plains, which is a big area of grasslands in
       the middle of the United States. The Cheyenne people were very good at riding horses and hunting animals like buffalo.'
Output:
{
  'definitions': ['A Native American tribe that lived in the Great Plains'],
  'category': 'One',
  'explanation': 'The response defines Cheyenne as a Native American tribe with kid-friendly details (Great Plains, horses,
        hunting). These are elaborations of one meaning, not distinct senses.'
}
```

Figure 6: System prompt used to extract definitions in the ELI5 prompting setting

Language	Type	Prompt Template
English	Normal	What is the definition of ‘{word}’?
	Simple	What is the definition of {word}’ in simple language?
	ELI5	Explain {word}’ like I am 5 years old.
	MSA	Keep in mind that some words have more than one meaning.
Russian	Normal	Каково определение понятия «{word}»?
	Simple	Каково определение слова «{word}» на простом языке?
	ELI5	Объясните «{word}» так, будто мне 5 лет.
	MSA	Пожалуйста, подумайте о различных контекстах или, если нужно, спросите.
Arabic	Normal	ما هو تعريف "{word}"؟
	Simple	ما هو تعريف "{word}" بلغة بسيطة؟
	ELI5	اشرح "{word}" وكأنني في الخامسة من عمري.
	MSA	يُرجى التفكير في السياقات المختلفة أو السؤال عن أحدها، إذا لزم الأمر.
Chinese	Normal	「{word}」的定义是什么？
	Simple	用通俗的语言来说，“{word}”是什么意思？
	ELI5	请像我只有五岁一样解释「{word}」。
	MSA	如果这个词有不同的语境，请考虑这些语境，或者询问我需要哪一种。
French	Normal	Quelle est la définition du terme « {word} » ?
	Simple	Quelle est la définition de « {word} » en langage simple ?
	ELI5	Expliquez-moi « {word} » comme si j’avais 5 ans.
	MSA	Pensez aux différents contextes ou demandez-en un, si nécessaire.

Table 8: Used prompt types across all languages. Multi-Sense-Aware (MSA) follows any of the other prompts.

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